Response to Reviewer 1 Comments

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Multi-Granularity Feature Decomposition with Ranking-Aware Regularization for Old Film Restoration

1. Summary

We thank Reviewer 1 for the careful and rigorous review. In this revision we have substantially strengthened the manuscript to address the six main concerns: (1) clarified the novelty of MGFE/RATR relative to the MambaOFR backbone and existing multi-scale / token-based / Mamba-based restoration methods; (2) adopted a single project-internal retraining protocol under the same computational budget; (3) froze measured ablations for fusion strategy, ranking metric, diversity loss, and granularity, with F1 (average fusion) as the main model; token insertion and branch aggregation are excluded until measured; (4) regenerated the qualitative figures with higher-resolution ROIs and added Supplementary Video S1; (5) froze the efficiency analysis by reusing prior 007 round2 L40 unified benchmark artifacts for baseline rows and newly verifying the F1 row on L40; and (6) added a dedicated least-improved failure-case panel and expanded limitations discussion.

2. Questions for General Evaluation

Reviewer's Evaluation	Response and Revisions
Does the introduction provide sufficient background and include all relevant references? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Introduction and Related Work were expanded with a focused novelty-positioning table and a clearer symmetry-scope argument.
Is the research design appropriate? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The research design was strengthened by adopting a single project-internal retraining protocol under the same computational budget for all compared variants.
Are the methods adequately described? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Methods section now includes the two formal symmetry properties, the insertion-operator details, and the hyperparameter-selection rationale.
Are the results clearly presented? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Results section now reports the extended ablation study, the complexity analysis, and the failure-case analysis.
Are the conclusions supported by the results? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Conclusions were tightened to reflect the F1 main-model results and the honest scope of the symmetry claim.
Are all figures and tables clear and well-presented? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. Figures 4 and 5 were regenerated with higher-resolution ROIs and three-class annotations; a new supplementary video was added.

3. Point-by-point response to Comments and Suggestions for Authors

Comments 1: The novelty over the existing MambaOFR backbone is not sufficiently justified. The manuscript should more clearly explain the theoretical and architectural distinction between the

proposed MGFE/RATR modules and existing multi-scale or token-based restoration frameworks, and provide a stronger comparison against recent Mamba-based restoration methods.

Response 1:

We thank the reviewer for this central concern and agree that a clearer positioning was needed. We added an architecture × theory difference table that distinguishes MgfrOFR from four families of methods (multi-scale pyramids, token-stripe ReID, Mamba-restoration, and diversity-regularization) along six dimensions: granularity source, branch diversity, invariance, feedback, inference cost, and target task. Relative to MambaOFR, the increment is granularity decomposition plus ranking-aware diversity regularization rather than a redesign of the scan mechanism. With the D2 model-selection decision, the performance claim is anchored to the F1 (MGFE+RATR, average fusion) main model: it improves over MambaOFR on both fidelity and perceptual metrics on the synthetic benchmark while keeping the overhead explicit in the new complexity table. The baseline complexity rows reuse prior 007 round2 L40 unified benchmark artifacts, and the F1 row was newly verified on L40 for this revision.

Revision made:

Added Table 1 (positioning table) and a difference paragraph in Related Work Section 2.3; aligned the Introduction and SSM novelty argument. The qualitative framing now shifts from a marginal PSNR chase to a validated combination of fidelity gain, perceptual improvement, and interpretable symmetry-based regularization.

Comments 2: The experimental validation lacks sufficient fairness and reproducibility details. Several compared methods (DeOldify, OldPhoto, DeepRemaster) are not originally designed for old film restoration, which may bias the comparison. The manuscript should provide clearer information regarding retraining procedures, hyperparameter tuning, degradation adaptation, and whether all baselines were optimized equally under the same computational budget.

Response 2:

We thank the reviewer for raising fairness and reproducibility. We adopt a single project-internal retraining protocol under the same pipeline and computational budget for all compared variants, directly addressing the fairness concern. Enhancement-oriented methods (DeOldify/OldPhoto/DeepRemaster) are explicitly labeled as reference methods rather than task-matched baselines. The reported main results now use the F1 model selected under this protocol.

Revision made:

Added a Baseline Adaptation Protocol paragraph and footnotes in the Results section; updated Tables 1 and 2 with F1 main-model numbers. Evidence boundary: current numeric table is frozen for measured entries; unmeasured entries are excluded.

Comments 3: The ablation study is incomplete and does not fully validate the contribution of each proposed component. The effect of the token insertion strategy, the choice of max fusion over average fusion, the influence of different ranking metrics, and alternative branch aggregation strategies and diversity regularization losses are not analyzed.

Response 3:

We thank the reviewer for requesting a more complete ablation. We now provide single-variable ablations: token insertion (I1, prepend), branch aggregation (AG1), fusion strategy (F1 average fusion as the main model versus A3 max fusion as the ablation control), ranking metric (R1), and diversity

loss (D1). The D2 decision record shows that A3 (max fusion) suffered from negative synergy on full-reference metrics, while F1 (average fusion) achieved the best synthetic fidelity-and-perception combination; we therefore present F1 as the validated default and A3 as the sensitivity-driven justification for choosing average fusion. Evidence boundary: R1/D1 are frozen where measured; I1/AG1 are excluded until three-dataset testing exists.

Revision made:

Added expanded ablation table(s) in the Results section and updated the Discussion to explain why MGFE recalibration alone is weaker than the full F1 model and how RATR with average fusion gives the strongest fidelity-perception balance.

Comments 4: The qualitative comparisons in Figures 4 and 5 are visually informative, but the enlarged regions are too small to clearly observe restoration differences, and several methods appear visually similar. The authors should provide higher-resolution ROI comparisons with clearer annotations, especially for scratch removal, texture recovery, and temporal consistency evaluation. A supplementary video comparison would further improve the evaluation.

Response 4:

We thank the reviewer for the constructive suggestions. We regenerated Figures 4 and 5 with higher-resolution ROIs and targeted annotations, and added Supplementary Video S1 with side-by-side real-world comparisons of the input, MambaOFR, and MgfrOFR(F1).

Revision made:

Updated Figures 4 and 5 and added Supplementary Video S1. The figure/video provenance is frozen in `t5-figures/evidence/revision_visual_manifest_20260630.json` and `Paper/symmetry/supplementary/SuppVideoS1_manifest.json`.

Comments 5: The computational complexity analysis is insufficient for a practical restoration framework. No detailed runtime, GPU memory consumption, parameter efficiency, or FPS comparison is provided against MambaOFR and transformer-based baselines.

Response 5:

We thank the reviewer for requesting a thorough complexity analysis. We now report parameters, FLOPs, latency, peak GPU memory, and FPS in Table 7. The baseline rows (DeepRemaster, RTN, RRTN, and MambaOFR) reuse the frozen 007 round2 L40 unified benchmark artifacts generated under the same executable BasicOFR protocol. The MgfrOFR(F1) row was newly verified for this revision on L40 using the current F1 configuration and reports 8.754M parameters, 3107.8G FLOPs per seven-frame clip, 488.5 ms/clip latency, 14.3 FPS, and 0.61 GB peak memory.

Revision made:

Updated Table 7 (complexity) in the Results section and froze the provenance matrix linking each row to its source artifact.

Comments 6: The manuscript would benefit from a stronger discussion of limitations and failure cases. The current discussion briefly mentions over-smoothing and flow-alignment errors, but no dedicated failure analysis is presented.

Response 6:

We thank the reviewer for highlighting the need for a dedicated failure analysis. We froze the failure-case selection from F1/A0 per-frame evidence within the least-improved REDS synthetic clips 029, 022, and 017, and added a focused failure-case panel together with an expanded Limitations section.

Revision made:

Added Figure 10 with the selected least-improved cases and signed error-difference maps. The selection and final figure hash are recorded in `t5-figures/evidence/failure_frame_selection_20260630.json` and `t5-figures/evidence/revision_visual_manifest_20260630.json`.

4. Response to Comments on the Quality of English Language

Point: The reviewer's assessment of the English language was: The English could be improved to more clearly express the research.

We thank the reviewer. After all quantitative values were finalized, the revised manuscript was carefully proofread for terminology consistency, repeated wording, and overly long sentences while preserving the frozen numbers.

5. Additional clarifications

We sincerely thank Reviewer 1 for the careful and rigorous review, which has substantially improved the manuscript. A marked-up version with all changes highlighted has been uploaded together with the revised manuscript.